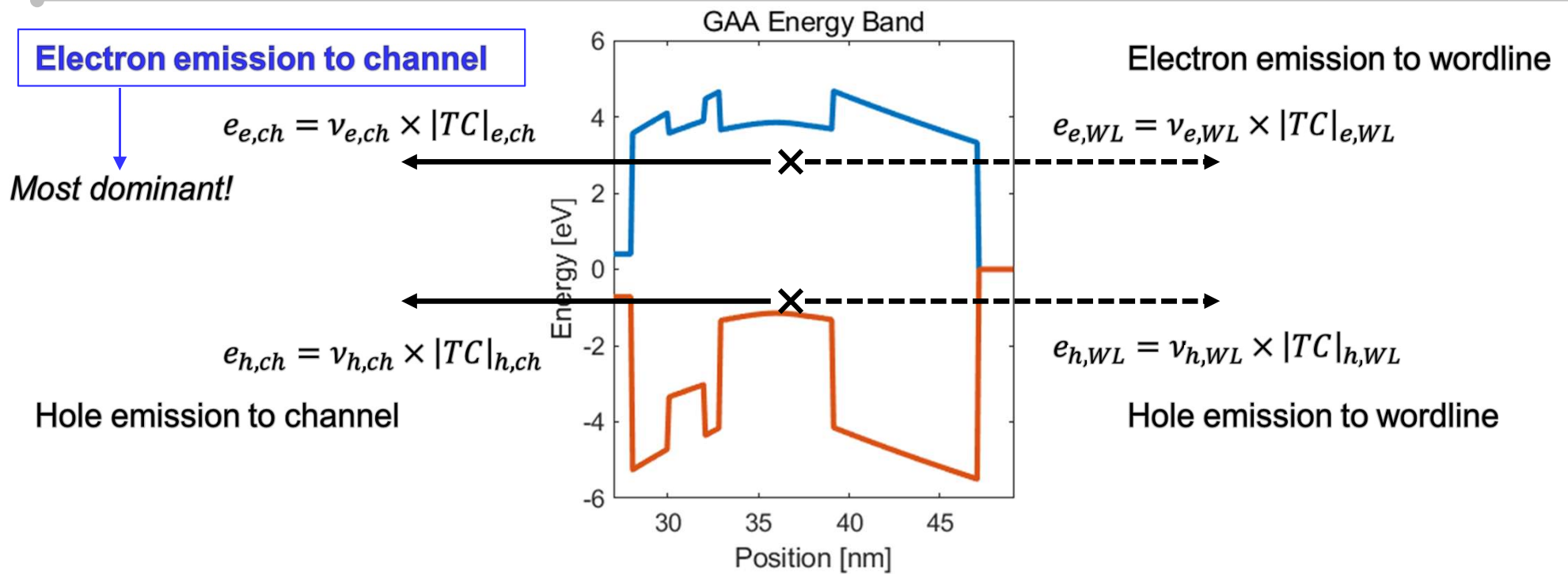


25년 12월 연구미팅자료

진명

Detrapping from CTN Overview



$$n_t = \sum_{i,j} n_{t,i,j} = \sum_{i,j} n_{t0,i,j} \{ \exp(-e_{ch,i,j} t) + \exp(-e_{WL,i,j} t) \}$$

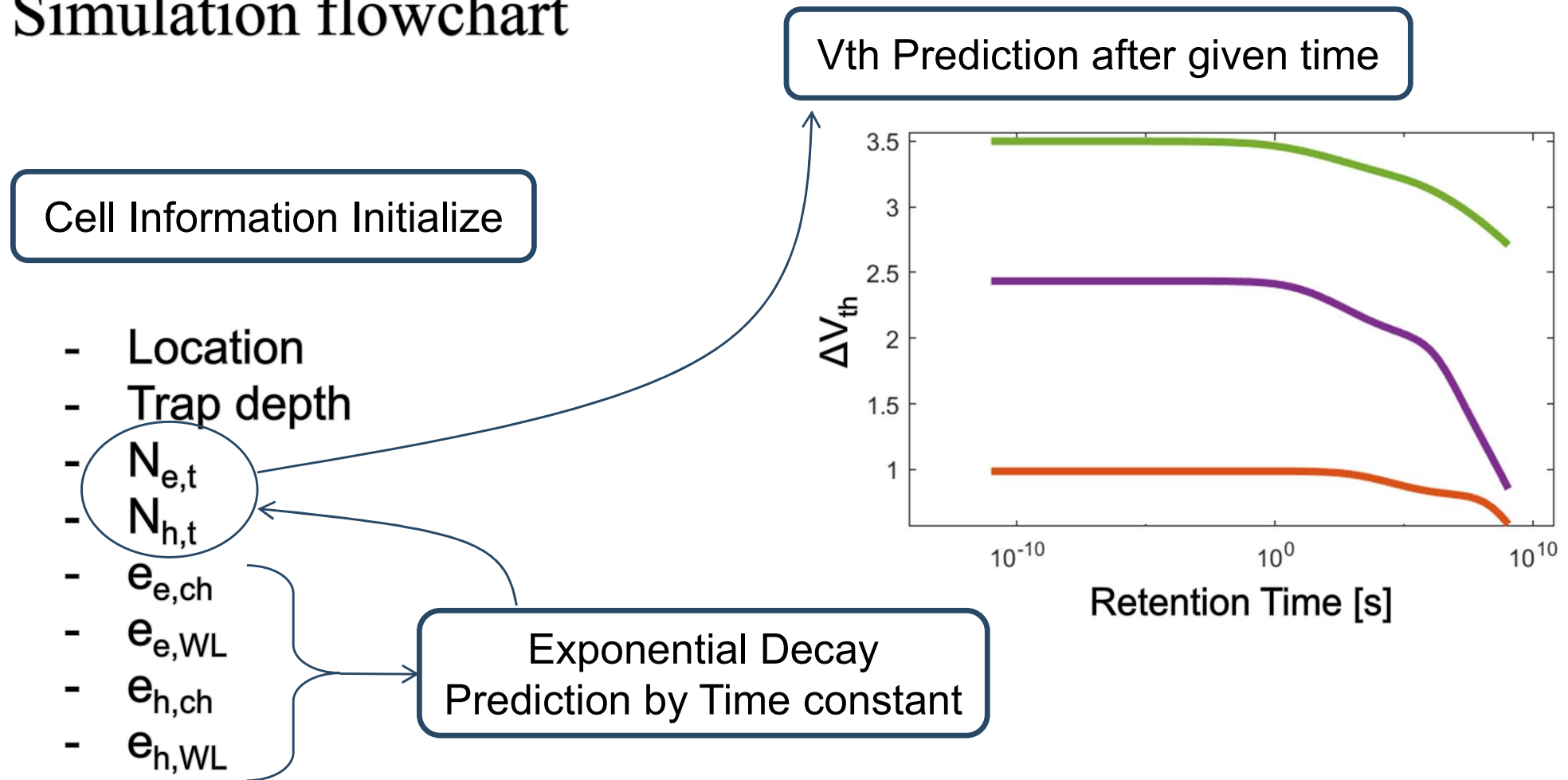
$$i_{ch} = -q \frac{dn_{t,ch}}{dt} = -q \frac{d}{dt} \sum_{i,j} n_{t0,i,j} \exp(-e_{ch,i,j} t) = q \sum_{i,j} e_{ch,i,j} n_{t0,i,j} \exp(-e_{ch,i,j} t)$$

$$i_G = -q \frac{dn_{t,G}}{dt} = -q \frac{d}{dt} \sum_{i,j} n_{t0,i,j} \exp(-e_{G,i,j} t) = q \sum_{i,j} e_{G,i,j} n_{t0,i,j} \exp(-e_{G,i,j} t)$$

=> Modeling by time transient & leakage current

Detrapping from CTN Overview

Simulation flowchart



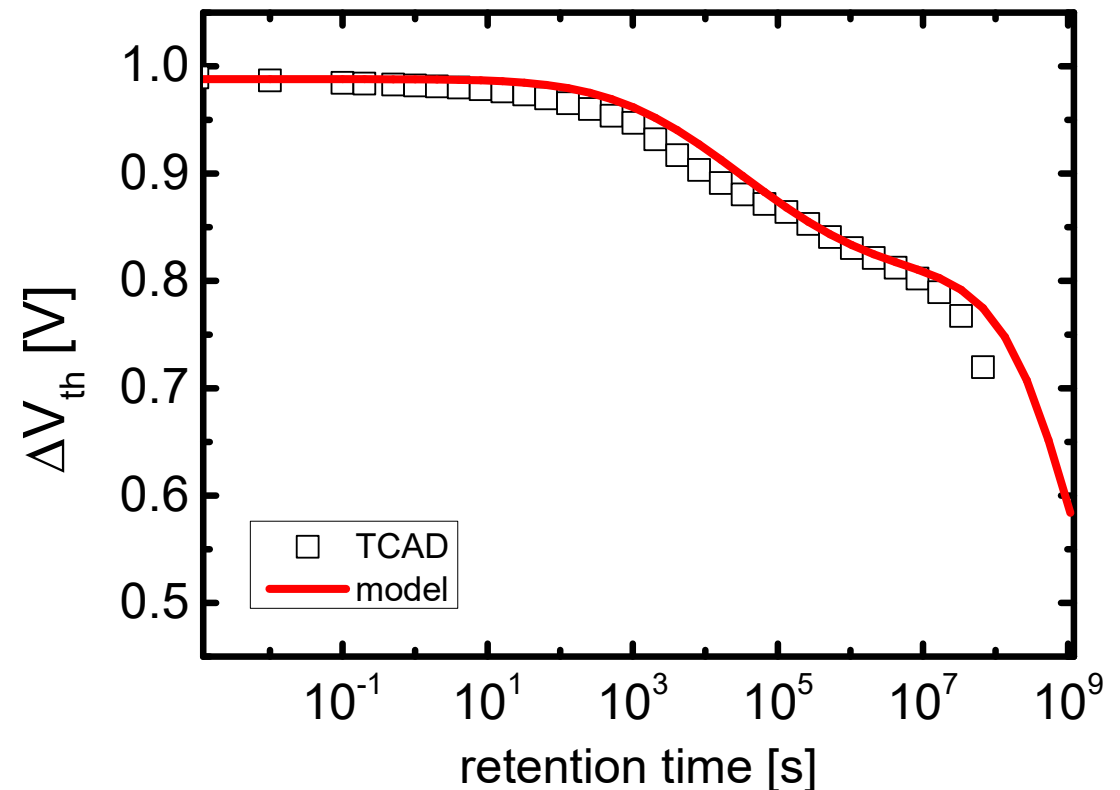
$$n_t = \sum_{i,j} n_{t,i,j} = \sum_{i,j} n_{t0,i,j} \{ \exp(-e_{ch,i,j} t) + \exp(-e_{WL,i,j} t) \}$$

Detrapping Transient & TCAD Fitting

Ongoing process for process variations – well match

Calibration condition

- 2/2/1/6/8 BETOX
- Shallow Trap: Gaussian
- Deep Trap: exact

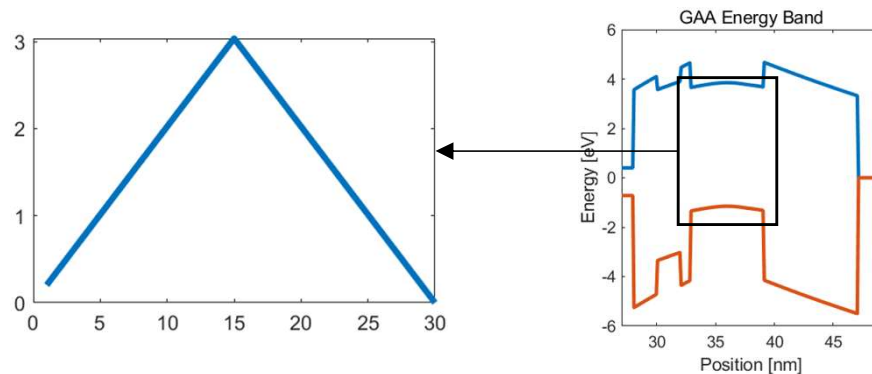


However, for simplicity of the research >> **ONO (5/6/8) Structure will be used for discussion**

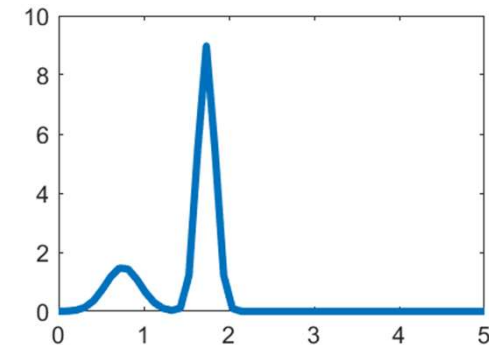
Detrapping Characteristic Variations

Variation Settings

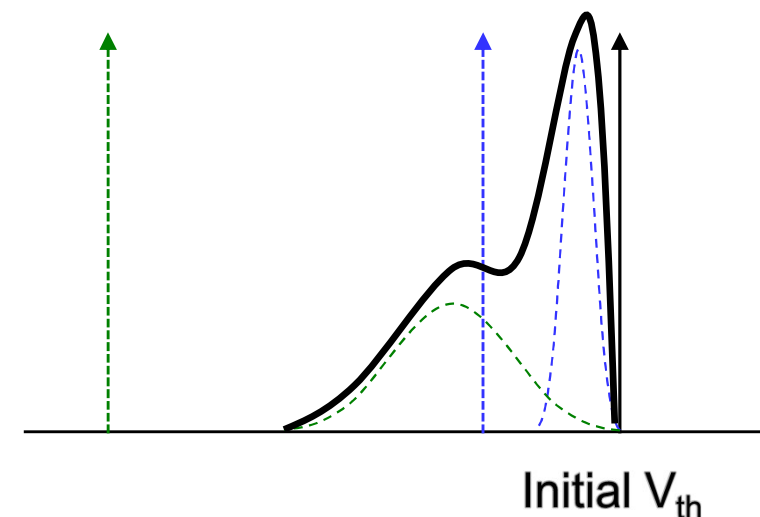
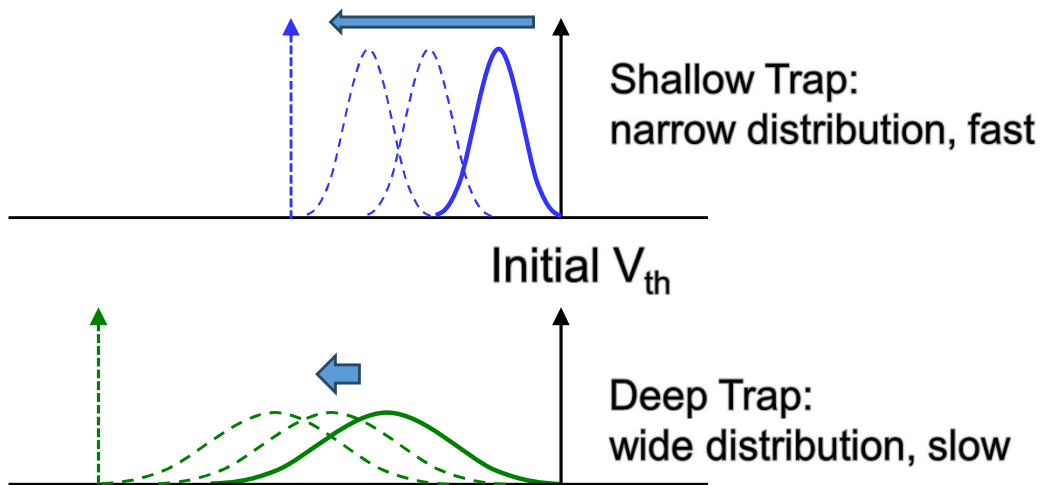
- CTN trapped electron profile



- Shallow/Deep trap profile

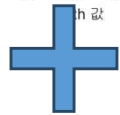
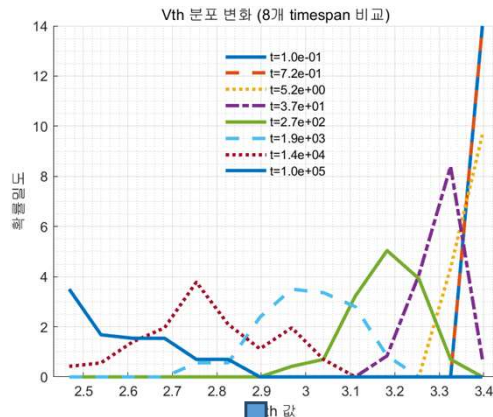


- Process Variation

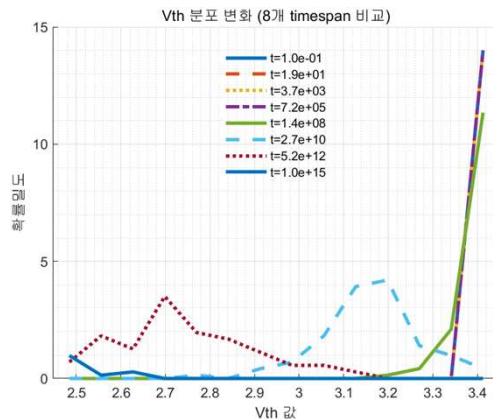


Detrapping Characteristic Variations

Only Shallow
(fast, narrow)



Only Deep
(slow, wide)



Vth 분포 변화 (8개 timespan 비교)

